

ERIC (CHEN-CHIEN) CHANG

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SUMMARY

- Backend / Data Infrastructure Engineer specializing in **performance tuning**, **distributed systems**, and **open-source data platforms**.
- Core stack: **Go**, **Java**, **Rust**, **TypeScript**, **PostgreSQL**, **Kafka**, **Docker**, **Kubernetes**. Delivered measurable wins including first-byte latency **100s to 5s**, WebUI load time **60s to 1s**, and simulation jobs **2hr to 1min**.

OPEN SOURCE CONTRIBUTIONS

Apache Kafka

2024 - Present

*Apache Kafka Contributor. Tech Stack: **Java**, **JVM***

[GitHub](#)

- Collaborating on Kafka Streams **KIP-770** input-buffer work ([PR #22458](#)) to add `input.buffer.max.bytes`, bounding buffering by bytes instead of record count for safer memory control under variable record sizes and partition assignments.
- Fixed **JVM container awareness** in producer performance tests and offset commit-before-partition-switch correctness ([PR #21584](#)).

Apache Gravitino

2024 - Present

*Apache Gravitino Committer. Tech Stack: **Java**, **Python**, **Docker**, **Kubernetes***

[GitHub](#)

- Added PostgreSQL/MySQL integration-test container management and reproducible Docker image workflows; reduced Hive image size by **420MB** and ported Python client partition APIs.

Apache DataFusion and DataFusion Comet

2025 - Present

*Contributor. Tech Stack: **Rust**, **Scala**, **Apache Spark***

[GitHub](#)

- Implemented Spark-compatible `str_to_map` in Rust ([PR #20120](#)) and wired Comet's native SparkStrToMap UDF with SQL file coverage ([PR #3654](#)).

redis/rueidis

Oct 2023 - Present

*Contributor. Tech Stack: **Go**, **Redis***

[PR #395](#)

- Implemented the go-redis API Adapter for teams migrating from go-redis.

SELECTED PROJECTS

TWFoundry

2026

*Real-time Taiwan transit data platform (hybrid cloud / homelab). Tech Stack: **Kafka**, **Flink**, **Iceberg**, **ClickHouse**, **Airflow**, **Kubernetes (k0s)**, **Cloudflare R2 / Workers / Pages**, **Docker**, **TypeScript***

[GitHub](#)

- Built a real-time transit data platform ingesting Taiwan TDX bus telemetry on a 5-minute cadence through **Kafka** into a partitioned **Cloudflare R2** data lake, running 24/7 on a self-hosted single-node **k0s Kubernetes** cluster.

- Architected a **Lambda** pipeline: a **Flink** speed layer for live anomaly detection (service-gap / bus-bunching) over the stream, plus an **Airflow**-orchestrated batch layer landing an **Apache Iceberg** lakehouse on R2 with **ClickHouse** OLAP modeling observations across service date, time slot, route, direction, vehicle, freshness, route progress, and stop context for headway / bunching and data-quality analysis.
- Served replayable map / timeline projection APIs from partitioned R2 artifacts via **Cloudflare Workers** / **Pages** Functions at the edge, decoupling public reads from the homelab cluster (no fallback on cache misses).

EXPERIENCE

Lawsnote

January 2024 - December 2024

Backend Software Engineer. Tech Stack: **Typescript, NodeJS, Docker, MongoDB, Prometheus, Grafana** Taipei, Taiwan

- Implemented **HTTP streaming** for CPU-intensive computation results, reducing first-byte latency from **100s** to **5s**.
- Built a **MinIO events** and **Elasticsearch** monitoring path to detect missing data proactively and improve pipeline reliability.
- Profiled and fixed a **NAPI-rs** performance issue with **Pyroscope**, improving Node.js native package execution.
- Built an **LLM service** for extracting information from ambiguous regulatory text and automated government act crawling, re-crawl, and re-parse workflows.

Mediatek (Contract)

February 2022 - March 2023

Backend Software Engineer. Tech Stack: **Go, Linux, Docker, PostgreSQL** Hsinchu, Taiwan

- Maintained a low-power simulation and benchmarking data warehouse, improving WebUI page load time from **60s** to **1s** with **PostgreSQL index-only scans**: [Blog Post](#).
- Used **Pyroscope eBPF** to identify import bottlenecks, then applied **PostgreSQL bulk import** to reduce simulation job duration from **2hr** to **1min**.
- Designed a message-queue-based microservice architecture and operated a **5-node** service cluster for horizontally scalable long-running jobs.
- Reduced Go service memory consumption by **50%** using **sync.Pool**: [Blog Post](#).
- Built integration-test fixture factories and maintained observability with **Prometheus, Grafana, OpenTelemetry**, and **Pyroscope**.

National Chen-Kung University

September 2017 - July 2019

Research Assistant. Tech Stack: **Python, C**

Tainan, Taiwan

- Conducted FDM 3D printing research across firmware, filament experiments, and hardware/software integration.
- Built experimental tooling with **Python, C, MicroPython**, and **Arduino**.

EDUCATION

National Taiwan Ocean University, Electrical Engineering coursework (Drop Out)

2013 - 2017

SKILLS

Languages: Go, Java, Rust, TypeScript, Python, C

Data Warehouse / Lakehouse: PostgreSQL, ClickHouse, Hive

Streaming / Processing: Kafka, Kafka Streams, Flink, Spark, SQL

Runtime / Orchestration: Airflow, Docker, Kubernetes, Linux

Observability / Data Quality: Prometheus, Grafana, OpenTelemetry, Pyroscope, Great Expectations

DevOps / Applied AI: Git, GitHub Actions, LLM service integration